System Capacity & Care Load Analytics for Unaccompanied Children

U.S. HHS Unaccompanied Children Program | Jan 2023 - Dec 2025
Python (cleaning, validation, EDA) + Power BI (2-page dashboard)

Project report by Aditaya Singh

1. Project Overview

This project looks at how children move through the U.S. HHS Unaccompanied Children Program: apprehended and placed in CBP custody, held in CBP custody, transferred out of CBP custody, cared for by HHS, and discharged from HHS care. The goal was to understand system capacity and care load over 2023 to 2025 and to turn the findings into clear business recommendations.

The work has two parts. In Python I cleaned and validated the data, explored it, and built monthly averages. In Power BI I built a two-page dashboard from the cleaned file.

Tools: Python, Pandas, NumPy, Matplotlib, Jupyter Notebook, Power BI.

Key numbers

Measure	Value
Rows x columns, raw file	1,170 x 6
Rows x columns, after cleaning	720 x 6 (450 fully empty rows removed)
Date range	12 Jan 2023 to 21 Dec 2025
Monthly periods analysed	36
Average children in HHS care	6,061 (daily min 1,972, max 11,516)
Average children in CBP custody	171.49 (max 531)
Average apprehensions	93.52 (max 333)
Average transfers out of CBP	128.67 (max 440)
Average HHS discharges	173.41 (max 505)

All averages are taken over the recorded observations, not over every calendar day (see Section 8).

2. Dataset

Source file: **HHS_Unaccompanied_Alien_Children_Program.csv**. It has six columns:

Column	Meaning
Date	Date of the observation
Children apprehended and placed in CBP custody*	Children coming into the system
Children in CBP custody	Children held by CBP on that date
Children transferred out of CBP custody	Children leaving CBP custody
Children in HHS Care	Children in HHS care on that date
Children discharged from HHS Care	Children leaving HHS care

3. Workflow

Load data, then clean it (empty rows, duplicates, date, HHS Care numbers), then validate it (negative and decimal values), then explore it (year, month, statistics, date gaps), then build monthly averages, then chart the trends in Python, then build the Power BI dashboard, then write findings and recommendations.

4. Python Work, Step by Step

Every step below comes from the notebook **UAC-System-Capacity-Analytics_Steps_Added_EN.ipynb**. Results are the outputs the notebook shows.

4.1 Load and inspect the data

This part loads the file and checks what it looks like before changing anything.

Step	What I did	Result
1	Imported pandas, numpy and matplotlib.	Libraries ready.
2	Loaded the HHS CSV file into **df** with **pd.read_csv**.	Dataset loaded.
3	Ran **df.head(10)** to see the structure.	Date is stored as text (for example, December 21, 2025). The other 5 columns are numeric counts.
4	Checked the shape with **df.shape**.	1,170 rows and 6 columns.
5	Printed the column names with **df.columns**.	Date plus the five measure columns listed in Section 2.

4.2 Data cleaning

This part removes empty rows and fixes the Date and HHS Care columns so they can be used for analysis.

Step	What I did	Result
6	Counted nulls in each column with **df.isnull().sum()**.	450 nulls in every column.
7	Selected rows where every column is null.	450 rows are completely empty.
8	Removed the fully empty rows with **dropna(how="all")**.	Empty rows removed.
9	Checked the shape again.	720 rows and 6 columns.
10	Ran **df.duplicated()**.	All visible values are False. I did not calculate the total with .sum(), so the exact count is not shown.
11	Checked the data type of Date.	Stored as a string.
12	Converted Date with **pd.to_datetime**.	Conversion done.
13	Checked the Date type again.	Now **datetime64**. The conversion worked.
14	Rechecked nulls.	0 nulls in every column.
15	Ran **df.info()**.	720 non-null values in every column. HHS Care was still text.
16	Looked at the HHS Care column.	Values are text with commas, for example "2,484".
17	Removed the commas with **str.replace(",", "")**.	Commas removed.
18	Converted the column with **pd.to_numeric**.	Conversion done.
19	Checked the HHS Care type again.	**int64**.

4.3 Data validation

Two quick checks that the counts make sense as numbers of children.

Step	What I did	Result
20	Looked for values below 0 in the apprehended column.	No rows found. No negative values.
21	Looked for decimal values in the transfers column.	No rows found. All values are whole numbers.

4.4 Exploratory data analysis (EDA)

This part checks the time coverage, the spread of the numbers, and how complete the dates are.

Step	What I did	Result
22	Found the earliest and latest dates.	12 Jan 2023 to 21 Dec 2025.
23-24	Created a **Year** column and counted it.	720 values.
25	Counted rows per year.	2024: 251, 2025: 239, 2023: 230.
26-27	Created a **Month** column and shortened it to Jan, Feb, and so on. This overwrote the first Month column.	Month column ready.
28	Displayed the table after adding Year and Month.	720 rows.
29	Counted rows per month.	April has the most (66). January has the fewest (52).
30	Ran **df.describe()**.	Apprehended: mean 93.5, max 333. CBP custody: mean 171.5, max 531. Transfers: mean 128.7, max 440. HHS Care: mean 6,061.3, min 1,972, max 11,516. Discharges: mean 173.4, max 505.
31	Calculated the difference between consecutive sorted dates.	Gaps between records calculated.
32	Counted how often each gap occurs.	1 day: 558. 3 days: 122. 4 days: 23. 2 days: 10. 5 days: 3. 6, 7 and 10 days: 1 each. The data is not recorded every day.

4.5 Monthly summary

The daily records are uneven, so I grouped them by month and took the average.

Step	What I did	Result
33	Created a **Year_Month** column from Date.	Monthly period column ready.
34	Built a monthly table with the mean of the 5 measure columns.	Table **monthly** created.
35	Viewed it with **monthly.head()**.	Jan 2023: apprehended 30.9, CBP custody 43.9, transfers 34.5, HHS Care 7,369.6, discharges 232.0.
36	Checked the shape.	36 rows and 6 columns, one row per month.

4.6 Charts in Python

Each chart answers one question. Steps 37, 39, 41, 43, 45 and 47 only rechecked the column names before the next chart.

Step	What I did	Result
38	**Chart 1: HHS care load.** Line chart of monthly average HHS care. Question: when did the load rise and fall?	About 7,400 in Jan 2023. Peak near 11,000 around Dec 2023. Down to about 6,000 during 2024. Sharp fall in Jan to Feb 2025, then flat at 2,000 to 2,500.
40	**Chart 2: CBP custody.** Line chart of monthly average CBP custody. Question: when was the pressure highest?	Peak near 440 in Jan 2024. Between 200 and 340 through 2024. After Dec 2024 it fell to 20 to 35 and stayed there.
42	**Chart 3: Apprehensions.** Line chart of monthly average apprehensions. Question: when did the incoming workload rise?	Peak around 225 in Jan 2024. Between 120 and 165 through 2024. Below 10 after Dec 2024.
44	**Chart 4: Transfers vs HHS discharges.** Both measures on one chart.	In 2023 discharges (200 to 350) were above transfers (90 to 210). In Jan to Mar 2024 transfers overtook discharges and peaked near 315. From mid 2024 to Jan 2025 both stayed between 150 and 200. After Feb 2025 both fell below 20.
46	**Chart 5: Apprehensions vs transfers.** Incoming children and transfers out on one chart.	Transfers stayed above apprehensions in most months and both lines follow the same pattern. After Jan 2025 both dropped to 5 to 10 and became almost equal.
48	**Chart 6: CBP custody vs HHS care.** Scatter chart of daily observations.	Two clusters. In one, CBP custody is below 60 and HHS care is 2,000 to 2,500. In the other, CBP custody is 100 to 500 and HHS care is 4,500 to 11,500. Above 100, higher CBP custody does not show a clear straight-line rise in HHS care. I did not calculate a correlation.

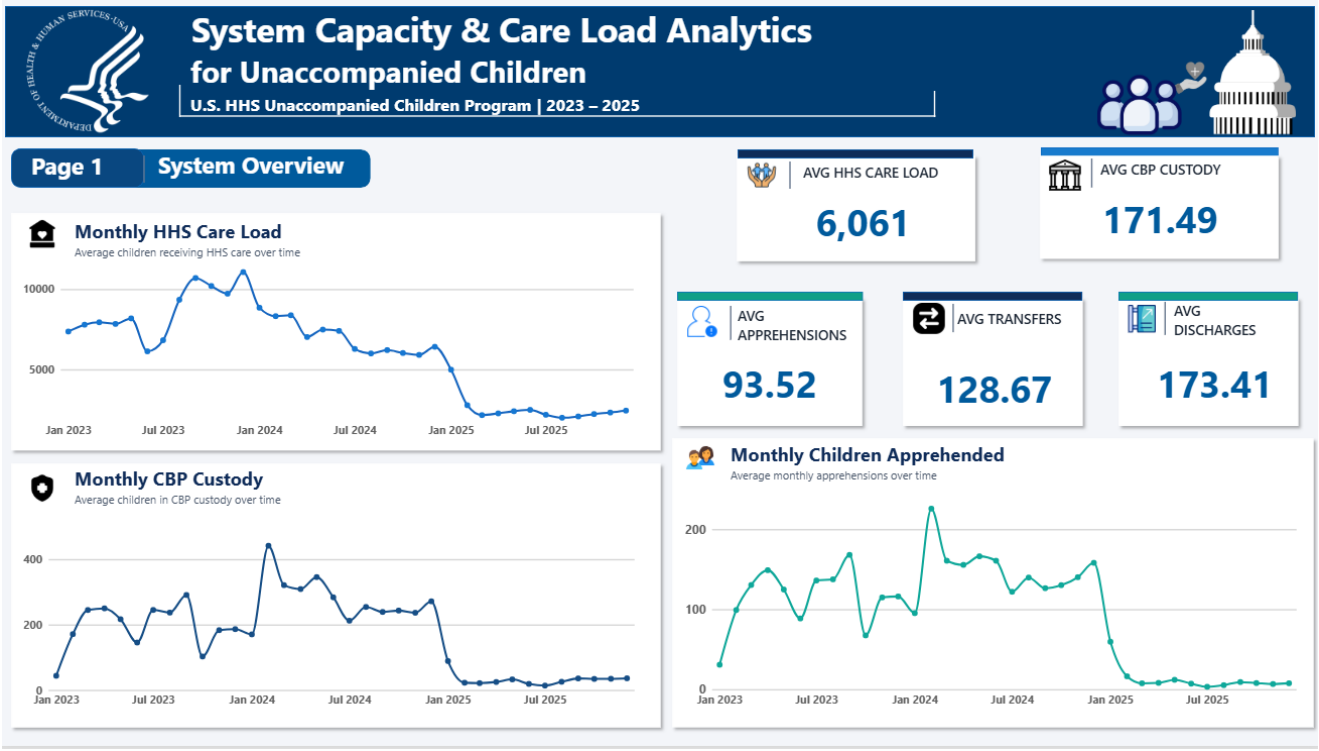
4.7 Save the cleaned data

Step	What I did	Result
49	Exported the cleaned **df** to **UAC_Cleaned_Data.csv** without the index.	Cleaned file saved. Power BI uses it as the table **UAC_Cleaned_Data**.

5. Power BI Dashboard

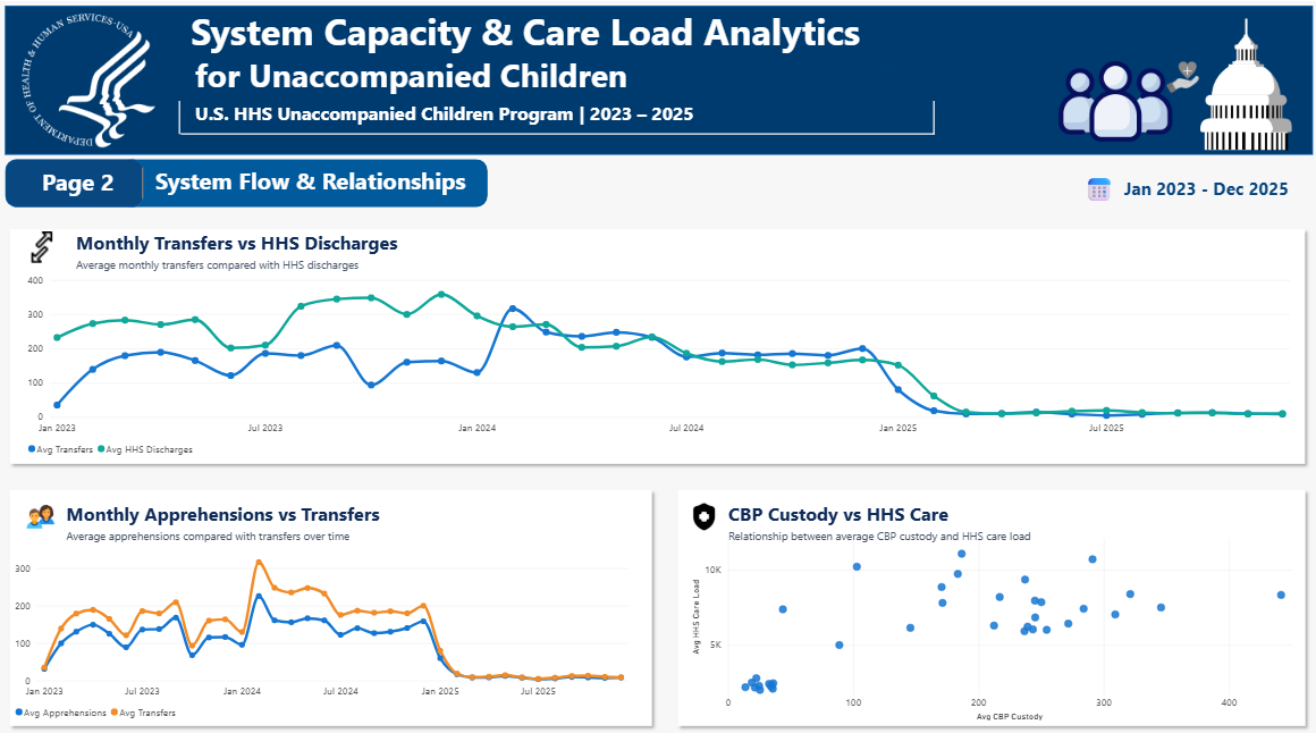
The cleaned file was loaded into Power BI. The report has two pages.

Page 1: System Overview



Visual	What it shows
5 KPI cards	Average HHS care load 6,061, CBP custody 171.49, apprehensions 93.52, transfers 128.67, discharges 173.41.
Monthly HHS Care Load	Rises from about 7,400 to a peak near 11,000 around the end of 2023, eases through 2024, then drops sharply in early 2025 to a low, flat level.
Monthly CBP Custody	Peaks near 440 in Jan 2024, stays between 200 and 340 through 2024, then drops to 20 to 35 from Jan 2025.
Monthly Children Apprehended	Peaks near 225 in Jan 2024, stays between 120 and 165 through 2024, then falls below 10 from Jan 2025.

Page 2: System Flow & Relationships



Visual	What it shows
Monthly Transfers vs HHS Discharges	Discharges run above transfers through most of 2023. Transfers jump past discharges in early 2024. The two lines then move together until both fall to near zero after Jan 2025.
Monthly Apprehensions vs Transfers	Transfers sit just above apprehensions in most months. The lines rise and fall together, then both drop to a very low level after Jan 2025.
CBP Custody vs HHS Care	Each dot is one month. A tight group at the bottom left is the low-volume period of 2025. The other months are spread out, with no clean straight line.

6. Key Findings

#	Finding	Evidence
1	**The system had three phases.**	A build-up through 2023, a high level through 2024, and a sharp drop from Jan 2025 that stayed low to Dec 2025. Apprehensions, CBP custody, transfers, discharges and HHS care all dropped together.
2	**Peak pressure came around the turn of 2023 to 2024.**	HHS care peaked near 11,000 (highest daily value 11,516). CBP custody peaked near 440 and apprehensions near 225 in Jan 2024.
3	**Care load moves slowly, intake moves fast.**	Monthly HHS care stays high for many months after apprehensions fall, and the early-2025 drop in HHS care came right after the drop in intake.
4	**Transfers follow apprehensions closely.**	Both lines have the same shape, with transfers above apprehensions in most months.
5	**CBP custody alone does not predict HHS care.**	The scatter shows two clusters and no clear straight-line pattern above 100 children in custody.
6	**The data is not daily.**	Gaps of 3 days appear 122 times, and the longest gap is 10 days.

7. Business Recommendations

These recommendations come from the patterns in this dataset. The data shows what happened, not why, so each one should be checked with program owners before it is acted on.

#	Recommendation	Why the data supports it
1	**Plan capacity for the peak, not the average.**	Average HHS care is 6,061, but the highest day is 11,516 and the standard deviation is 2,833. Plan beds, staff and budget around peak levels with a buffer, and review them monthly.
2	**Set early-warning triggers on intake.**	Apprehensions and CBP custody both peaked in Jan 2024 (about 225 and 440 on a monthly average). Track a rolling 7-day and 30-day average of apprehensions and raise an alert when it crosses an agreed level, so surge capacity opens before CBP custody builds up.
3	**Keep transfers at or above the pace of intake.**	Transfers stayed above apprehensions in most months, which kept CBP custody from piling up. Track the transfer-to-apprehension ratio each week and act when it falls below 1.
4	**Do not treat 2025 as the new normal.**	From Jan 2025 every measure fell to a low, flat level (HHS care 2,000 to 2,500, CBP custody 20 to 35, apprehensions under 10). Keep a standby surge plan so the system can scale up again, and confirm with the data owner whether the drop reflects real demand or a change in policy or reporting.
5	**Do not forecast HHS care from CBP custody alone.**	The scatter shows no clear straight-line link above 100 children in custody. Build any forecast with more inputs, such as transfers, discharges and time lags. Calculate the correlation first.
6	**Check the 2023 flow numbers with the data owner.**	In 2023 monthly discharges were above transfers, yet HHS care still rose to about 11,000. A closed flow would not do this. HHS may receive children from sources that are not in this file, or the timing may differ. Resolve this before using the numbers for staffing or budget.
7	**Report every day and store counts as numbers.**	122 gaps of 3 days and one gap of 10 days make monthly averages less reliable. HHS Care also arrived as text with commas. Ask for daily records and clean numeric fields at the source.

8. Limitations and Next Steps

Item	Note
Averages, not totals	Averages use the recorded observations only. The Jan 2023 HHS Care figure of 7,369.6 is the average of the days recorded in January, not the level on every day.
Duplicates	The notebook shows **df.duplicated()** output without a total. Adding **.sum()** would confirm the exact count.
Correlation	The scatter is read by eye. Calculating the correlation (and a lagged one) would back up Finding 5.
Causes	The dataset has no policy, border or funding information, so it cannot explain why the 2025 drop happened.
Next steps	Add length-of-stay data, add a forecast for HHS care, and add a daily or weekly view in Power BI.

9. Project Files

File	Purpose
HHS_Unaccompanied_Alien_Children_Program.csv	Raw dataset
UAC-System-Capacity-Analytics_Steps_Added_EN.ipynb	Python notebook: cleaning, validation, EDA, charts
UAC_Cleaned_Data.csv	Cleaned data used in Power BI
Power BI report (.pbix)	Two-page dashboard